Final Bioinformatics Report

Analysis of *Escherichia coli* K-12 MG1655 Reference Genome

Internship	SWYNEX Technologies
Dataset	*Escherichia coli* K-12 MG1655 Complete Reference Genome
NCBI Accession	NC_000913.3
Tools	Python, Google Colab, Matplotlib

1. Research Question

What are the basic sequence characteristics and nucleotide composition of the *E. coli* K-12 MG1655 reference genome, and what can these characteristics tell us about the selected biological dataset?

2. Objective

To clean a publicly available biological sequence, calculate basic nucleotide statistics, visualize the results, and interpret the findings using computational bioinformatics methods.

3. Dataset

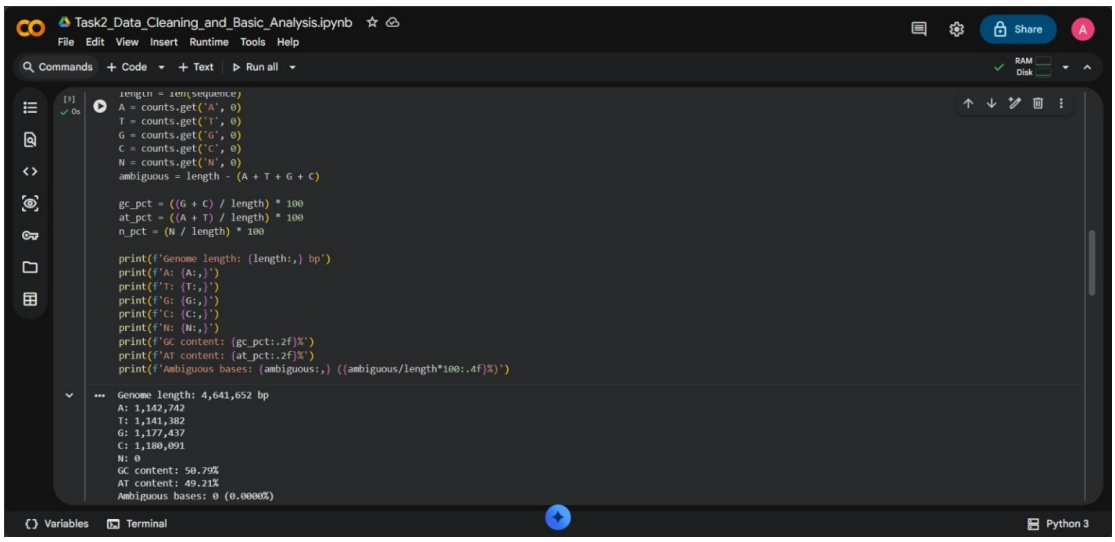
The selected dataset is the complete reference genome of *E. coli* K-12 MG1655 obtained from the NCBI database.

Genome length: 4,641,652 bp
Accession: NC_000913.3
Sequence type: Complete reference genome
Molecule: Circular chromosome

4. Methods

The FASTA sequence was retrieved from NCBI and processed using Python in Google Colab. The workflow included FASTA sequence retrieval, removal of formatting and whitespace, conversion to uppercase, nucleotide counting, GC and AT percentage calculation, quality checking for ambiguous bases, and visualization using Matplotlib.

Task 2: Nucleotide statistics



5. Results

Parameter	Result
Genome length	4,641,652 bp
A	1,142,742
T	1,141,382
G	1,177,437
C	1,180,091
GC content	50.79%
AT content	49.21%
Ambiguous bases	0

Interpretation

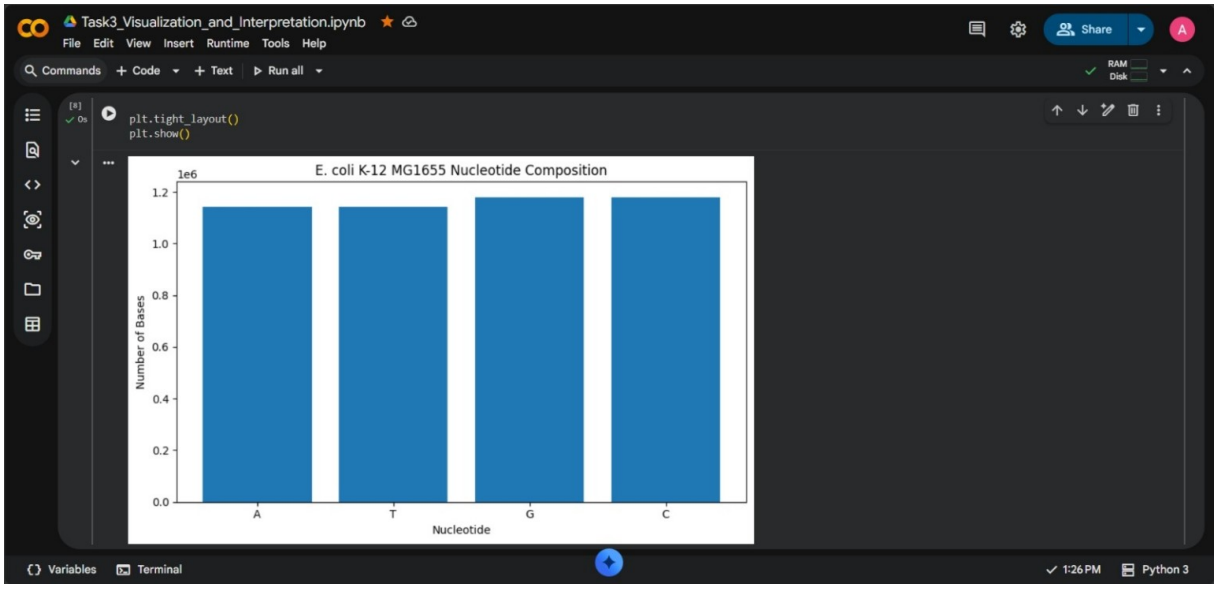
G and C were slightly more abundant than A and T. The genome showed a nearly balanced overall composition, with 50.79% GC and 49.21% AT.

The absence of ambiguous N bases indicates that the cleaned reference sequence contained no unknown nucleotide positions.

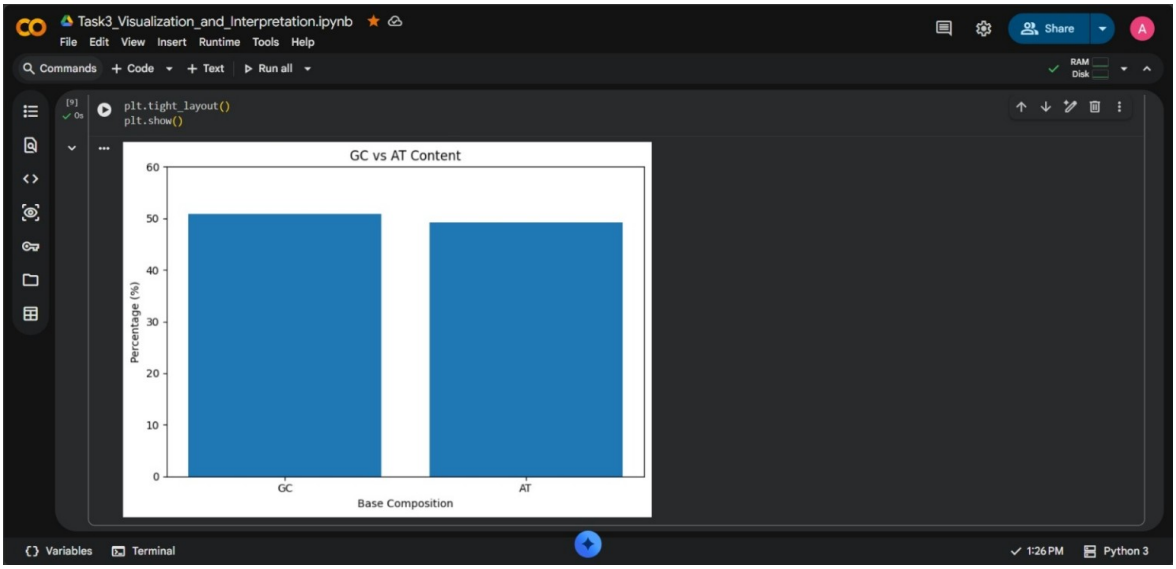
6. Visualization

The nucleotide composition and GC/AT content were visualized using Python and Matplotlib. The visualizations show that G and C occur slightly more frequently than A and T, GC and AT content are close to an even 50:50 distribution, and the overall sequence contains no ambiguous bases.

Nucleotide composition



GC versus AT content



7. What the Results Do and Do Not Prove

The analysis describes the overall nucleotide composition and basic sequence quality of the selected genome. However, these results do not prove gene or protein function, evolutionary relationships, regulatory activity, biological mechanisms, or the importance of specific genomic regions. Additional genomic and functional analyses would be required to investigate these questions.

8. Limitations

Only one reference genome was analyzed. The analysis focused on basic sequence statistics. Overall GC content does not show local GC variation. No gene prediction or functional annotation was performed, and no comparative or phylogenetic analysis was performed.

9. Conclusion

The *E. coli* K-12 MG1655 reference genome was successfully cleaned, analyzed, and visualized using Python-based bioinformatics methods. The genome is 4,641,652 bp long and has 50.79% GC content, 49.21% AT content, and 0 ambiguous bases. The project demonstrates a basic workflow for transforming publicly available biological sequence data into meaningful statistical and visual summaries.

10. References

National Center for Biotechnology Information (NCBI). *E. coli* K-12 MG1655 reference genome, NC_000913.3.

Python Software Foundation. Python documentation.

Matplotlib Documentation. Python visualization library.